INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

PUMPS, CENTRIFUGAL, ELECTRIC-MOTOR-DRIVEN

(FOR WATER)

This specification is approved for use within the USA Belvoir Research, Development and Engineer Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers close-coupled, electric-motor-driven, horizontally mounted, centrifugal water preps complete with all necessary accessories and connections.

1.2 Classification. Pumps shall be of the following types and classes, as specified (see 6.2):

Type I - Integral self-priming.
  Class A - 200 gallons per minute (gpm) at 50-foot total head.

Type II Straight centrifugal
  Class B 25 gpm at 115-foot total head.
  Class C 750 gpm at 50-foot total head.
  Class D 70 gpm at 165-foot total head.
  Class E 125 gpm at 165-foot total head.
  Class F 250 gpm at 165-foot total head.
  Class G 500 gpm at 165-foot total head.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

FF-B-171	- Bearings, Ball, Annular (General Purpuse).

MILITARY

MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-F-10603	- Pumps and Pumping Units, Centrifugal, Power-Driven, for Water, Packaging of.
MIL-I-45208	- Inspection System Requirements.

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-838	- Lubrication of Military Equipment.
MIL-STD-889	- Dissimilar Metals.
MIL-SRD-1472	- Human Engineering Criteria for Military Systems, Equipment, and Facilities.
MIL-STD-1474	- Noise Limits for Army Materiel.

HANDBOOKS

MILITARY

MIL-HDBK-113	- Guide for the Selection of Lubricants, Fluids, Preservatives, and Specialty Produdcts for Use in Ground Equipment Systems.

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(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111 -5094.)

2.1.2 Other Government documnets, drawings, and Publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

OCCUPATIONAL SAFETY AND HEALTH ACT OF 1970 (OSHA)

Title 29, CFR, Chapter SVII, Part 1910, Subpart S - Electrical.

CODE OF FEDERAL REGULATIONS

Federal Food Drug and cosmetic, Food Additives Title 21, Part 177, 2600, Chapter I.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. When indicated reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

MILITARY TRAFFIC MANAGEMENT COMMAND (MIMC/TEA)

Pamphlet 70-1 - Transportability for Better Strategic Mobility.

(Application for copies should be addressed to the Military Traffic Management Command Transportation Engineer Agency (MIMC/TEA), ATTN: MIT-TR, P.O. Box 6276, Newport News, VA 23606-0276.)

2.2 NonGovernment publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the elicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B16.1 - Cast-Iron Pipe Flanges and Flanged Fittings.
B16. 14 - Ferrous Pipe Plugs, Bushings, and Locknuts with Pipe Threads.

(Applicationfor copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code Section IX, Welding and Brazing Qualifications.

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(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 48 - Gray Iron casting's.
A 582 - FREE Machining stainless and Heat-Resisting Steel Bars, Hot-Rolled or cold-Finished.
A 827 – Plates, Carbon, Steel for Forging and Similar Applications.
B 584 - Copper Alloy Sand Castings for General Applications.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY (AWS)

D1.1 Structural Welding Code, Steel.

(Application for copies should be addressed to the American Welding Society, 550 N.W. LeJeune Road, P.O. Box 351040, Miami, FL 33135.)

HYDRAULIC INSTITUTE (HI)

Hydraulic Institute Standard

(Application for copies should be addressed to the Hydraulic Institute, 1230 Keith Building, Cleveland, OH 44115.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

MG1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street NW, Washington DC 20037.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

1020 - Plain Carbon Steels, Group II, Case Hardening Grade.
J429 - Mechanical and Material Requirements for Externally Threaded Fasteners.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence . Nothing in this document, however supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3.

3.1 Description. The electric-motor-driven centrifugal pumps, hereinafter referred to as "pumping unit", shall consist of a centrifugal pump, electric rotor, and all necessary connections for water-pumping operations.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. All surfaces that contact potable water shall conform to the Federal Food, Drug and Cosmetic, Food Additives, Title 21, Code of Federal regulations for contact with food. The compounds listed in section 177.1020, 177.1030, 177.1040, 177.1050, or 177.1480 of the Federal Food, Drug and Cosmetic, Indirect Food Additives, Chapter 1, Title 21, Code of Federal Regulations shall not be acceptable materials, Questions pertinent to this effort shall be referred by the procuring activity to the departmental medical service who will act as advisor to the procuring agency.

3.2.1 Material deterioration prevention and control. The pumping unit shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operation and storage environment to which the pumping unit may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and subcomponents, and shall make the information available, upon request, to the contracting officer or designated representative.

3.3.4 Recovered materials. For the of this requirement , recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the pumping unit may be newly fabricated from recovered materials to the maximum extent practicable, provided the pumping unit produced meets all other requirements of this specification. Used, rebuilt or manufactured components, pieces and parts shall not be incorporated in the pumping unit.

3. 3.5 Hydrocarbon compatibility. Those components, parts, or aaccessories not specifically described shall be compatible with liquid hydrocarbon products including aviation gasolines, jet aircraft engine fuels, fuel oils, kerosene diesel oils.

3.4 Design and construction.

3.4.1 Screw threads. Screw threads shall conform to FED-STD-H28.

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3.4.2 Pump. The pump shall be a close-coupled, single-stage, single-suction, vertical-split-case, centrigugal pump. Each pump shall be provided with a pump-case drain port and plug. The drain port shall be internally threaded NPT and shall be of such size and so located as to completely drain the pump case in not more than 2 minutes. The drain plug shall be a square-head pipe plug conforming to ANSI B16.14. When the plug is in the volute passage, the plug shall be flush with the inner surface of the pump case.

3.4.3 Pump case and cover. The pump case and cover shall be iron castings conforming to ASTM A 48, class optional. The pump case or cover shall be provided with replaceable wear rings or plates, as applicable. The wear rings shall be bronze conforming to ASTM B 584, alloy UNS C83600. The wear plates shall be steel conforming to ASTM A 827 and SAE 1020.

3 .4.4 Pump impeller. The pump impeller shall be cast iron conforming to ASTM A 48, class 30, and shall be single-suction. The impeller shall be mounted on the extended rotor shaft and shall be keyed to the shaft and held in place by a lock-cap nut with a shaft-protecting washer. The lock-cap nut shall be screwed on the shaft in such a manner as to tighten as the rotor shaft rotates.

3.4.5 Pump-shaft seal. The pump shall be provided with a mechanical self-lubricant seal. The seal shall be lubricated by the liquid pumped. The seal shall be self-aligning and shall be removable from the pump shaft without disassembling the pump from the rotor mounting face. The adjustment of the sealing surface shall be independent of the impeller adjustment. All components of the seal shall be resistant to corrosion.

3.4.6 Pump-shaft sleeve. A shaft sleeve shall be provided between the pump shaft and the seal . The sleeve shall be corrosion-resistant steel conforming to ASTM A 582, type 416 (UNS S41600) .

3.4.7 Electric motors. The electric rotors shall conform to NEMA standard MG1, polyphase, squirrel-cage induction, solid-shaft, totally enclosed, fan-cooled, horizontally mounted design B, type C, face-mounted. The rating of the motors shall be based upon a 55 °C temperature rise. Insulating material of the motors shall be class H. Motors shall be provided with either a watertight terminal-connection compartment integral with the rotor frame or a watertight terminal box complete with a watertight stuffing-tube connector.

3.4.7.1 Power rating. The continuous-duty rating of the motor shall be not less than the required pump brake horsepower at rated pump output conditions (see 1.2).

3.4.7.2 Synchronous speed. The synchronous speed of the motor shall be 3,600 revolutions per minute (rpm) when connected to a 60-Hertz (Hz) power source.

3.4.7.3 Voltage and frequency. Unless otherwise specified (see 6.2), the rotor shall operate on a 208,/416 volt, 3-phase, 60-Hz, 4-wire electrical system.

3.4.7.4 Motor shaft. The rotor shall be provided with a steel shaft. The shaft shall support the maximum combined radial and thrust loads imposed by the rotating elements and hydraulic action of the pump.

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3.4.7.5 Motor bearings. The rotor shall be provided with antifriction, lubricated-for-life, sealed bearings conforming to FF-B-171. The bearings shall function on a continuous-duty basis while supporting the maximum, combined radial and thrust loads imposed by the rotating elements and hydraulic action of the pump.

3.4.8 Magnetic starters. When specified. (see 6.2), the motors shall have magnetic starters attached in waterproof enclosures. The magnetic starters shall be capable of starting the rotor when under the full load conditions imposed by the attached pump. The magnetic starters shall be unfused with separate overheat protectors in each phase. Spare overheat protector and solenoid elements as required shall be furnished so that the magnetic starter may be adapted to either 50-Hz or 60-Hz electrical supply.

3.4.8.1 Remote starters. When specified (see 6.2), a set of pushbuttons with pilot lights shall be furnished for mounting at a remote operating station.

3.4.9 Human factors engineering. The pumping unit shall conform to human factors engneering design criteria as described in MIL-STD-1472. Special emphasis shall be given to the following: 4.1; 4.2; 4.3; 4.3; 4.4.b, c, f, h, m, and p; 4.5; 4.6; 4.7; 4.8; 4.9; 4.11; 5.1; 5.2; all of 5.4 except for 5.4.1.1.2, 5.4.1.2.4, 5.4.1.8.5, 5.4.1.8.6, 5.4.2.2.2.6, 5.4.2.2.4, 5.4.2.2.5, 5.4.3.1.2, 5.4.3.1.3, 5.4.3.2.2, 5.4.3.2.3, 5.4.3.2.4, 5.4.3.2.5, 5.4.3.2.6, 5.4.3.2.7, 5.4.3.2.8, 5.4.5, and 5.4.6; 5.5; 5,6; 5.9; and all of 5.13 except for 5.13.2.3, 5.13.2.4, 5.13.2.5, 5.13.4.1, 5.13.4.2, 5.13.4.3, 5.13.7.1.8, 5.13.7.2.2, 5.13.7.4.2, and 5.13.7.5.

3.4.10 Noise limits. The noise produced by the pumping unit shall conform with MIL-STD-1474 requirements, with the exception of MIL-STD-1474, 5.2, 5.3, and 5.4, when tested in accordance with 4.5.2.3.5. The previsions of MIL-STD-1474, 4.3 and 4.4 shall be provided if and only if MIL-STD-1474, 5.1.1.2 procedures have been pursued and documented to the satisfaction of the procuring activity and written permission to exceed the 85 dB(A) limit is obtained from the procuring activity. Hazard signs shall conform with MIKL-STD-1474, 4.3 except the sign shall state "HEARING PROTECTION REQUIRED WITHIN (specify) FEET". The sign shall be readable at the distance indicated on the sign.

3.5 Maintainability.

3.5.1 Ease of maintenance. All assemblies, parts, and attachments shall be accessible for maintenance, repair and replacement without the removal of other major assemblies and installed attachments not normally removed. Drain outlets shall be located for accessibility and allow for drainage into a suitable container. Design for maintainability shall be in accordance with 3.4.9.

3.5.2 lubricants. The procedure for the selection of lubricants shall be in accordance with MIL-STD-838, section 5. Lubricants selected shall be in accordance with MIL-HDBK-113, chapter 2. When the specification of the lubricant selected includes a requirement for a Qualified Products List (QPL), the lubricant supplied shall be from a source that is listed on the applicable QPL.

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3.6 Transportability.

3.6.1 Tiedown provisions. The tiedown provisions shall conform to MIL-STD-209. The tiedown provisions shall satisfactorily complete the pull testing as specified in 4.5.2.5 without weld failure, permanent deformation, cracking, loosening, or breaking of the provision or its connecting structural component.

3.6.2 Slinging provisions. The slinging provisions shall conform to MIL-STD-209. The provisions shall enable the complete pump to be lifted in the normal operating position. The previsions shall be located so that not less than 1-inch clearance is maintained between slings and all exterior parts and shall be fastened to members which will withstand stresses in the amount and direction of pull specified for the provisions without weld failure, permanent deformation, cracking, loosening, or breaking of the provision or its connecting structural components. Slinging provisions may also be used as tiedown provisions when such provisions meet the requirements specified in 3.6.1. All slinging/tiedown provisions shall be labeled “LIFT”, "TIEDOWN", or "LIFT/TIEDOWN", as applicable, in 1-inch (2.54 cm) high letters.

3.6.3 Rail transportability. The pump shall be rail transportable in CONUS and NATO countries without restrictions. The pump shall have a dimensional profile within the Gabarit International de Chargement (GIC) in accordance with MIMC/TEA pamphlet 70-1, outline diagram when loaded on a 50-inch (127 cm) high rail car. The pump shall be capable of withstanding shock loads resulting from rail impact testing in accordance with 4 .5.2.7 without failure, damage, or permanent deformation. A transportability report is required (see 6.7).

3.7 Performance.

3.7.1 Hydrostatic pressure. The pump shall withstand an internal hydrostatic pressure of not less than 1.5 times the shutoff head for a period of not less than 3 minutes without leakage or permanent deformation.

3.7.2 Impeller balancing. The impeller shall be dynamically balanced. The dynamic "out-of-balance" condition shall be not more than the following:

0.025 ounce-inch for class A, B, D, and E pumps
0.050 ounce-inch for class C, F, and G pumps.

3.7.3 Type I, integral self-priming. The pump shall be an integral self-priming centrifugal pump (after the pump case has been initially filled).

3.7.3.1 Class A pump. The pump shall be provided with a semi-open impeller. The suction and discharge ports of the pump shall be internally threaded 2-inch NPT. The discharge port shall be vertical and shall be located at the top of the pump case. The pump shall deliver not less than 200 gpm at 50-foot total head when operated at a speed of not more than 3,600 rpm and using a 10-foot static Suction lift.

3.7.3.1.1 Check valve. A check valve shall be provided on the suction inlet of each pump. When subjected to a maximum pressure equal to the shutoff head of

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the pump, the valve shall prevent siphoning of water from the pump case and shall not leak or show permanent deformation of the body, cover, seat, or any of its components.

3.7.3.1.2 Priming. When operating at conditions equivalent to standard sea level atmospheric conditions (sea level atmospheric conditions of 29.92 inches of mercury and water with a specific weight of 62.3 pounds per cubic foot at a temperature of 68 °F), the pump shall prime in not more than 2 minutes under a 10-foot, static suction lift. The suction and discharge fittings shall be of the same nominal size as the suction and discharge ports of the pump.

3.7.4 Type II, straight centrifugal. The pump shall be a straight centrifugal pump and shall be provided with an enclosed impeller. The pump stroll be provided with suction and discharge flanges conforming to ANSI B16. 1, class as specified herein. The pump discharge port shall be capable of being located in any one of four positions (vertical upward, horizontal to the left, vertical downward and horizontal to the right). The pump case shall be provided with drain ports and plugs. The ports and plugs shall be as specified in 3.4.2. The drain ports shall be located so as to completely drain the pump case in not more than 2 minutes when the discharge port is mounted in any one of its four positions.

3.7.4.1 Class B pump. The pump shall be provided with a 1-1/2 inch flanged suction and a 1-1/4-inch flanged discharge. The pump suction and discharge flanges shall be cast integrally with the pump case or rover, as applicable. The discharge opening shall be positioned vertically upward andd shall be on the right side of the pump case when viewed from the pump end. The pump shall deliver not less than 25 gpm at 115-foot total head when operated at a speed not more than 3,600 rpm and using a suction lift equivalent to 10 feet.

3.7.4.2 Class C pump. The pump shall be provided with a 5-inch flanged suction connection and a 4-inch flanged discharge connection. The pump suction and discharge flanges shall be cast integrally with the pump case or rover, as applicable. when viewed from the pump end, the discharge flange shall be located at the top of the pump case and shall be horizontally to the left. The pump shall deliver not less than 750 gpm at 50-foot total head when operated at a synchronous speed of not more than 3,600 rpm and using a static suction lift equivalent to 10 feet.

3.7.4.3 Class D pump. The pump shall be provided with 1-1/2 inch screwed or integrally cast flanges rated to exceed the maximum pressure of the pump. When viewed from the pump end, the discharge flange shall be located at the top and discharge liquid to the left. The pump, when connected to a 60Hz power source, shall deliver not less than 70 gpm at a total head of 165 feet with a suction lift equivalent to 10 feet.

3.7.4.4 Class E pump. The pump shall be provided with 2-inch screwed or integrally cast flanges rated to exceed the maximum pressure of the pump. When viewed from the pump end, the discharge flange shall be located at the top and discharge liquid to the left. The pump shall deliver not less than 125 gpm at a total head of 165 feet with a suction lift equivalent to 10 feet.

3.7.4.5 Class F pump. The pump shall be provided with 4-inch screwed or integrally cast flanges rated to exceed the maximum pressure of the pump. when viewed from the pump end, the discharge flange shall be located at the top and

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discharge liquid to the left. The pump shall deliver not less than 250 gpm at a total head of 165 feet with a suction lift equivalent to 10 feet.

3.7.4.6 Class G pump. The pump shall be provided with 4-inch screwed or integrally cast flanges rated to exceed the maximum pressure of the pump. When viewed from the pump end, the discharge flange shall be located at the top and discharge liquid to the left. The pump shall deliver not less than 500 gpm at a total head of 165 feet with a suction lift equivalent to 10 feet.

3.8 Safety. All rotating, reciprocating, exposed, and electrically energized parts that are of such nature or located so as to become a hazard to operating personnel shall be fully guarded or enclosed in accordance with OSHA 29CFR1910.309. Protective devices shall not impair the operating functions of the pumping unit.

3.9 Color and finish. The portions of the pump normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G, color as specified (see 6.2).

3.10 Identification marking. The pump shall be identified in accordance with MIL-STD-130. The marking shall be applied to the pump on plates conforming to MIL-P-514, type I, style 1, composition C, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.11 Workmanship.

3.11.1 Castings and forgings. All parts, components, and assemblies of the pumping unit which include castings and forgings shall be clean of harmful extraneous material such as sand, dirt, sprues, scale, and flux. Rework shall be limited to procedures which do not reduce mechanical properties or affect function.

3.11.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material Shall be done by methds that will not cause damage to the material. corners shall be square and true. Flame-cutting using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges, and corners, except where sharp edges or corners are required or where they are not detrimental to safety.

3.11.3 Welders and welding.

3.11.3.1 Welders. Before assigning any welder or welding operator to manual welding work covered by this specification, the contractor shall obtain certification that the welder or welding operator has passed qualification tests as prescribed by either the ASME Boiler and Pressure Code - Steel or AWS D1.1 for the type of welding operations to be performed and that such qualification is effective as defined by the particular code. Certification shall be made available, upon request, to the contracting officer or designatd representative. Contractors who make horizontal welds need not qualify welders for "all position welding".

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3.11.3.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping or wire brushing, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 3/16 inch unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication of the pumping unit shall be at the option of the contractor.

3.11.4 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing hits conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement. Bolts shall protrude not more than two full threads.

2.11.5 Machine work. Tolerances and gages for metal fits shall conform to the limits specified herein.

4. QUALITY ASSURNACE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards as applicable

4.1.3 Inspection system. The contractor must maintain and inspection system in accordance with MIL-I-45208.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (See 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article shall be tested as specified 4 .5.2 through 4.5.2.5, as applicable. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1.

4.4.3 Tests.

4.4.3.1 Individual. Each pump shall be tested as specified in 4.5.2.1 and 4.5.2.2. Failure of either test shall be cause for rejection.

4.4.3.2 Samples. Samples selected in accordance with 4.4.1 shall be tested as specified in 4 .5.2.3.1 for type I pumps and as specified in 4.5.2 .3.2 for both type I and type II pumps.

4.5 Inspection procedure.

4.5.1 Examination. The pumping unit shall be examined as specified herein for the following defects:

Major

101. Materials not as specified (see 3.3).
102. Materials not resistant to corrosion aned deterioration, or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments (see 3.3.1).
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.2).
104. Contractor does not have documentation available for identification of material, material finishes, or treatment (see 3.3.3).
105. Used, rebuilt or remanufactured components, or parts incorporate in the truck (see 3.3.4).
106. Hydrocarbon compatibility not as specdified (see 3.3.5).
107. Screw threads not as specified (see 3.4.1).
108. Pump not as specified (see 3.4.2).
109. Pump case and rover not as specified (see 3.4.3).
110. Pump impeller not as specified (see 3.4.4).

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111. Pump-shaft seal not as specified (see 3.4.5).
112. Pump-shaft sleeve not as specified (see 3.4.6).
113. Electric motors not as specified (see 3.4.7).
114. Power rating not as specified (see 3.4.7.1).
115. Synchronous speed not as specified (see 3.4.7.2).
116. Voltage and frequency not as specified (see 3.4.7.3).
117. Motor shaft not as specified (see 3.4.7.4).
118. Motor bearings not as specified (see 3.4.7.5).
119. Magnetic starters not as specified (see 3.4.8).
120. Remote starters not as specified (see 3.4.8.1).
121. Human factors engineering not as specified (see 3.4.9).
122. Ease of maintenance not as specified (see 3.5.1).
123. Lubricants not as specified (see 3.5.2).
124. Tiedown previsions not as specified (see 3.6.1).
125. Slinging previsions not as specified (see 3.6.2).
126. Rail transportability not as specified (see 3.6.3).
127. Hydrostatic pressure not as specified (see 3.7.1).
128. Impeller balance not as specified (see 3.7.2).
129. Type I, integral self-priming pump not as specified (see 3.7.3).
130. Class A pump not as specified (see 3.7.3.1).
131. Check valve not as specified (see 3.7.3.1.1).
132. Priming not as specified (see 3.7.3.1.2).
133. Type II, straight centrifugal pump not as specified (see 3.7.4).
134. Class B pump not as specified (see 3.7.4.1).
135. Class C pump not as specified (see 3.7.4.2).
136. Class D pump not as specified (see 3.7.4.3).
137. Class E pump not as specified (see 3.7.4.4).
138. Class F pump not as specified (see 3.7.4.5).
139. Class G pump not as specified (see 3.7.4.6).
140. Safety not as specified (see 3.8).
141. Castings and forgings not as specified (see 3.11.1)
142. Metal fabrication not as specified (see 3.11.2).
143. Welders and welding not as specified (see 3.11.3).
144. Welders certificate not as specified (see 3.11.3.1).
145. Bolted connections not as specified (see 3.11.4).
146. Machine work nut as specified (see 3.11.5).

Minor

201. Color and finish not as specified (see 3.9).
202. Identification marking not as specified (see 3.10).

4.5.2 Tests. Perform tests in accordance with the applicable previsions of the test code for centrifugal pumps of the Hydraulic Institute Standard. Conduct tests using water as the test liquid and correct all test data to standard sea level atmospheric conditions of 29.92 inches of mercury and to the specific weight of water (62.3 pounds per cubic foot) at a temperature of 68 °F.

4.5.2.1 Pump and pump-shaft seal hydrostatic pressure. With the pumpshaft seal installed, subject the pump and pump-shaft seal to an internal hydrostatic pressure of not less than 1.5 tires the shutoff head of the pump for not less than 3 minutes. Leakage or permanent defamation of the pump or pump-shaft seal or nonconformance to 3.7.1 shall constitute failure of this test.

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4.5.2.2 Pump impeller balancing. During assembly of the pump, determine the dynamic "out-of-balance" condition of the impeller. A dynamic “out-of-balance” condition more than that specified in 3.7.2 shall constitute failure of this test.

4.5.2.3 Performance.

4.5.2.3.1 Priming. Prime and operate the type I pump in accordance with the conditions specified in 3.7.3.1.2. Priming in more than 2 minutes shall constitute failure of this test.

4.5.2.3.2 Head capacity. Operate the pump to determine the head capacity under the conditions specified for the designated class as applicable. The operating time shall be that required to obtain the performance characteristics of the pump. Nonconformance to the head and capacity for the designated class, as applicable, shall constitute failure of this test.

4.5.2.3.3 Horsepower requirements. Operate the pump as specified in 4.5.2.3.2 for not less than 8 hours. During the last hour of the test, determine the horsepower required by the pump. Horsepower required by the pump in excess of the continuous-duty horsepower of the motor shall constitute failure of this test.

4.5.2.3.4 Check valve. After the type I pump has been primed as specified in 3.7.3.1.2, subject the check valve to a pressure (total static discharge head) of not less than the shutoff head of the pump. Shut off the pump and allow it to sit for not less than 30 minutes. Restart the pump. Determine whether the pump delivers water immediately. Immediate delivery of water is an indication that the valve prevented siphoning of water from the pump case. Failure of the pump to maintain prime, leakage, or permanent defamation of any component of the valve shall constitute failure of this test.

4.5.2.3.5 Noise level test. Noise levels shall be measured in accordance with MIL-STD-1474 requirements and Reported in the format indicated by MIL-STD-1474, figure 7. As a minimum, noise levels shall be measured when equipment is operating under full load. MIL-STD-1474, 5.1.2.1.4 contours shall be taken at not fewer than 12 equal (horizontal) arc increments, one incurment shall include data from the noisiest position. Additionally, the noise level at the typical operating position shall be provided as dB(A) level. Failure to comply with MIKL-STD-1474 shall constitute failure of this test.

4.5.2.4 Pump-case drain.

4.5.2.4.1 Type I pump. With the pump case filled with water, remove the drain plug to determine the drainage time of the pump case. Failure of the pump case to completely drain in 2 minutes or less shall constitute failure of this test.

4 .5.2.4.2 Type II pump. With the pump case filled with water and with the discharge port mounted in any one of its four positions (vertical up, horizontal to the left, vertical down, and horizontal to the right), remove the drain plug to determine the drainage time of the pump case. Repeat the same operation for

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each of the four positions of the discharge port. Failure of the pump case to completely drain, with the discharge port in any of its four positions, in 2 minutes or less shall constitute failure of this test.

4.5.2.5 Tiedown provisions test. The tiedown provisions shall be tested in accordance with MIL-STD-209 to prove conformance to 3.6.1. Inability to meet the requirements of 3.6.1 shall constitute failure of this test.

4.5.2.6 Slinging provisions test. The slinging provisions shall be tested in accordance with MIL-STD-209 to prove conformance to 3.6.2. Inability to meet the requirements of 3.6.2 shall constitute failure of this test.

4.5.2.7 Rail impact test. The first article pump shall be tested in accordance with MIL-STD-810 to prove conformance to 3.6.3. Inability to meet the requirements of 3.6.3 shall constitute failure of this test.

4.6 Inspection of packaging. The preservation, packing and marking shall be examined and tested to determine compliance with the applicable quality assurance provisions of MIL-P-10603 and the requirements of section 5 herein.

5. PACKAGING

5.1 First article pack. Unless otherwise specified (see 6.2), the contractor shall furnish a first article pack for examination and test, within the time frame specified (see 6.2), in accordance with the pumpproduction pack requirements of MIL-P-10603.

5.2 Preservation, packing and marking. Preservation, packing and marking shall be in accordance with level A, B or C requirements of MIL-P-10603.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The pumps are intended for use in water purification units and plants, laundry facilities and for use in general-purpose, water-pumping operation.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Type and class of pump required (see 1.2).
d. When a first article is required for inspection and approval, number of unit required (see 3.2).
e. Voltage or frequency when other than specified (see 3.4.7.3).
f. When magnetic starters are required (see 3.4.8).
g. When remote starters are required (see 3.4.8.1).
h. Color and finish required (see 3.9).

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i. When the welding process will not be at the option of the contractor (see 3.11.3.2).
When a first article pack is not required (see 5.1) .
k. Time frame for submission of first article pack, when required (see 5.1).
l. Level of preservation, packing and marking required (see 5.2).
m. When a transportability report is required (see 6.7).
n. Specification part number required (see 6.8).

6.3 First article. When a first article inspection is required, the item should be a preproduction model. The first article should consist of one units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the frost article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Lubricants. MIL-STD-838 Lubrication of Military Equipment, prescribes the policy for using specif ication-type products wherever possible and provides specific requirements for potential use of non-standard proprietary products. MIL-STD-838 is implemented by MIL-HDBK-113 Guide for the Selection of lubricants, Fluids, Preservatives and Specialty Products for Use in Ground Equipment systems. The contracting officer should note that unless otherwise authorized by the US Army Belvoir Research, Development and Engineering Center (ATTN: STRBE-VF), Fort Belvoir, VA 22060-5606, lubricants, fluids, and greases for ground equipment systems must be restricted to those listed under MIL-HDBK-113, chapter 2.

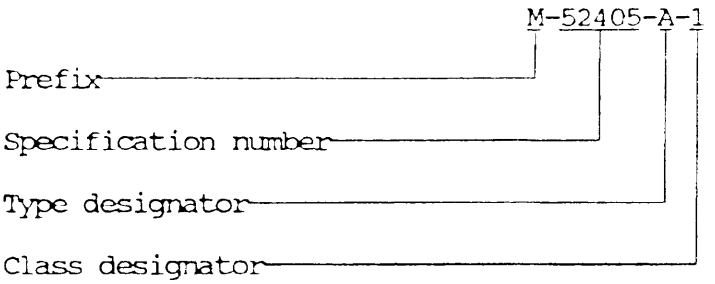
6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools, as necessary (including any special tools), and instructions regarding shipment of pump units.

6.6 First article pack. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preserve pack and mark the pumps and accessories in accordance with this specification

6.7 Transportability report. When specified (see 6.2), the contracting officer should arrange for the contractor to furnish a transportability report in accordance with AR 70-47 and S.O.P. 70-38 for the pump, 95 days prior to delivery of the first article pump.

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6.8 Part or identifying number (PIN). The PIN to be used for acquired to this specification are created as follws:



6.8.1 Type designator. A one position field used to designate the required type (see table I).

TABLE I. Type.

Letter designator	Type designation
A	I - Integral self-priming
B	II - Straight centrifugal

6.8.2 Class designator. A one position field used to designate the required class (see table II).

TABLE II. Class.

Number designator	Class designation
1	A-200 gpm at 50-foot total head
2	B-25 gpm at 115-foot total head
3	C-750 gpm at 50-foot total head
4	D-70 gpm at 165-foot total head
5	E-125 gpm at 165-foot total head
6	F-250 gpm at 165-foot total head
7	G-500 gpm at 165-foot total head

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6.9 Subject term (key word) listing.

Close-coupled
Horizontally mounted
Integral self-priming
single-stage
Single-suction
Straight centrifugal
Vertical-split-case

6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME

Review activities:
Army - MD, GL
DLA - CS

User activity:
Navy - YD)

Preparing activity:
Army – ME

Project 4320-A232

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-52405C	901022

3. DOCUMENT TITLE

Pumps, Centrifugal, Electric-Motor-Driven (for Water)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	e. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
	(703) 664-5717 354-5717
c. ADDRESS (Include Zip Code) US Army Belvoir RDE Center ATTN: STRBE-TSE Ft. Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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